

BUYER AGENT — INDEPENDENT VERIFICATION REPORT

Formal Verification Audit for Branch: [feature/buyer-agent-verification](#)

Audit Date:	September 15, 2026 (02:26 IST)	Verification Branch:	feature/buyer-agent-verification
Current Commit:	fa8f9ed4ec2a9437a1eb661ce49e4d480a386858	PR Base Commit:	bc53986 ('farmer') — 4 commits ahead, 0 behind
PR Remote URL:	https://github.com/Bhaveshransan/FarmGenAI/pull/new/feature/buyer-agent-verification	Working Tree:	100% Clean (0 uncommitted modifications)
FarmerAgent Files:	100% UNTOUCHED (0 diff lines)	Audit Mandate:	Zero new code, zero synthetic data, zero assumptions

1. EXECUTIVE SUMMARY

This independent verification report audits all technical claims made regarding the BuyerAgent implementation on the AgriNegotiator repository. Every single claim was verified by directly executing Python inspectors against files, git trees, model binaries, and test suites without modifying repository code or retraining models.

Core Verification Results:

- Git Integrity (VERIFIED):** The verification branch feature/buyer-agent-verification is based on commit fa8f9ed, which sits 4 clean linear commits ahead of the teammate's base commit bc53986 ('farmer'). FarmerAgent source code (agents/farmer_agent.py) and tests are **100% untouched** (0 diff lines).
- 7-Crop Configuration & Isolation (VERIFIED):** Exactly 7 canonical Maharashtra crops are configured in shared/crop_catalog.py. 20 crop aliases normalize with 100% accuracy in \$O(1)\$ time. 9 tested unsupported crops (Tomato, Wheat, Potato, Maize, Cabbage, etc.) were confirmed to be strictly rejected by BuyerAgent with ValueError/REJECT.
- Dataset Counts & Integrity (VERIFIED DIRECTLY):** Direct CSV parsing proved the exact row counts: **62,429 raw records** in Monthly_data_cmo.csv (100% Maharashtra), **14,078 clean records** in clean_buyer_market_data.csv (0 duplicates, 0 nulls), and **13,179 feature rows** in buyer_feature_dataset.csv.
- Data Provenance (VERIFIED AS THIRD-PARTY MIRROR):** The raw APMC data originates from the Maharashtra State Agricultural Marketing Board (MSAMB / CMO Maharashtra open data), but was retrieved from a public GitHub mirror repository (ashushaw04/APMC_Argo_Challenge), not directly from the live data.gov.in API.
- 19,200 Record Claims (FALSE / CONTRADICTED):** There is **NO 19,200 record dataset** in the repository for either BuyerAgent (which has 14,078 real rows) or FarmerAgent (which has only 150 simulated records in historical_negotiations.json).
- ML Model Artifact & Metrics (VERIFIED ON DISK, NOT USED AT RUNTIME):** The trained Ridge Regression pipeline exists at backend/models/buyer_price_prediction_model.pkl (1,807 bytes). All reported test metrics (Persistence MAE 1.7258, Ridge MAE 2.1812, Moving Avg MAE 2.3949) were **100% reproduced mathematically**. However, BuyerAgent.make_offer() in agents/buyer_agent.py does **NOT CURRENTLY INVOKE** this model at runtime.

2. GIT & REPOSITORY EVIDENCE

Item	Verified Value	Verification Command / Evidence
Current Branch	feature/buyer-agent-verification	git branch (*)
Base Branch Commit	bc53986 ('farmer')	Ancestor commit on origin/main
Current Commit Hash	fa8f9ed4ec2a9437a1eb661ce49e4d480a386858	HEAD commit in git log
BuyerAgent Commits	4 commits: fa8f9ed, 000510d, b2a7157, d28d7c2	git log bc53986..HEAD --oneline
Working Tree Status	Clean (no uncommitted tracked modifications)	git status
FarmerAgent Diff	0 files changed, 0 insertions, 0 deletions (100% UNTOUCHED)	git diff bc53986..HEAD --agents/farmer_agent.py
FarmerAgent Tests Diff	0 files changed, 0 insertions, 0 deletions (100% UNTOUCHED)	git diff bc53986..HEAD --tests/test_05_farmer_agent_extensive.py
Remote Tracking	bhavesht/feature/buyer-agent-verification	git push -u bhavesht feature/buyer-agent-verification

3. BUYER AGENT CODE VERIFICATION

Component	Source File	Status	Verified Behavior
Class Definition	agents/buyer_agent.py	VERIFIED	class BuyerAgent(BaseAgent): line 68
Buyer Personas	agents/buyer_agent.py	VERIFIED	4 personas (retail_supermarket, bulk_wholesaler, food_processor, restaurant_kitchen)
Quality Utility Modulation	agents/buyer_agent.py	VERIFIED	calculate_utility() modulates utility by quality grade A/B/C and persona
Budget Guardrail (Inst 25)	agents/buyer_agent.py	VERIFIED	When affordable_qty <= 0, triggers immediate REJECT (no clamping to 1)
BATNA Refactor (Inst 26)	agents/buyer_agent.py	VERIFIED	Evaluates supplier alternatives or tags MARKET_REFERENCE_HEURISTIC
ZOPA Refactor (Inst 27)	agents/buyer_agent.py	VERIFIED	Checks budget ceiling without fabricating seller reservation price
Purchase Order Generation	agents/buyer_agent.py	VERIFIED	generate_purchase_order() emits canonical TitleCase crop term sheet
Offer Validation	agents/buyer_agent.py	VERIFIED	_validate_offer_inputs() intercepts non-positive prices, budget overflow, bad crops
Runtime ML Invocation	agents/buyer_agent.py	NOT IMPLEMENTED	Model exists on disk, but make_offer() does NOT call the model

4. DIRECT DATASET VERIFICATION (CALCULATED FROM DISK)

All figures calculated directly by opening and iterating every row in the files:

Metric / Attribute	Raw Dataset (Monthly_data_cmo.csv)	Clean Dataset (clean_buyer_market_data.csv)	Feature Dataset (buyer_feature_dataset.csv)
Exact File Path	backend/dataset/Monthly_data_cmo.csv	backend/dataset/clean_buyer_market_data.csv	backend/dataset/buyer_feature_dataset.csv
File Size	4,885,990 bytes (4.66 MB)	954,458 bytes (932 KB)	1,562,115 bytes (1.49 MB)
Verified Row Count	62,429 rows (MATCHES CLAIM)	14,078 rows (MATCHES CLAIM)	13,179 rows (MATCHES CLAIM)
Column Count	11 columns	10 columns	18 columns
Date Range	2014-09 to 2016-11 (27 months)	2014-09 to 2016-11 (27 months)	2014-09 to 2016-10 (26 months)
Maharashtra Records	62,429 (100.0% Maharashtra)	14,078 (100.0% Maharashtra)	13,179 (100.0% Maharashtra)
Target 7-Crop Total	14,317 raw rows	14,078 clean rows	13,179 feature rows
Bajra Count	2,346 raw	2,336 clean	2,198 feature rows
Cotton Count	1,063 raw	1,063 clean	941 feature rows
Jowar Count	3,716 raw	3,710 clean	3,491 feature rows
Onion Count	1,872 raw	1,867 clean	1,771 feature rows
Rice / Paddy Count	1,580 raw	1,363 clean (217 dups merged)	1,269 feature rows
Soybean Count	3,727 raw	3,726 clean	3,499 feature rows
Sugarcane Count	13 raw	13 clean	10 feature rows
Duplicate Keys	42 exact raw commodity duplicates	0 duplicate keys	0 duplicate keys
Null / Empty Values	0 empty cells	0 empty cells	0 empty cells

5. DATA PROVENANCE & REAL VS SYNTHETIC VERIFICATION

- **Origin Classification:** THIRD-PARTY MIRROR OF OFFICIAL MAHARASHTRA GOVERNMENT APMC DATA.
- **Retrieval Source URL:** https://raw.githubusercontent.com/ashushaw04/APMC_Argo_Challenge/master/Data/Monthly_data_cmo.csv.
- **Government Authority:** The dataset contains authentic public open data published by the Maharashtra State Agricultural Marketing Board (MSAMB / CMO Maharashtra) covering 349 APMC mandis. However, it was ingested from a public GitHub mirror, not a live authenticated query to data.gov.in.
- **Authenticity Verification (REAL):** Inspection of individual observations confirmed authentic spot prices, real APMC committee names, and characteristic reporting anomalies (e.g. 86 records with inverted min/max prices). No synthetic generation scripts exist for market data.
- **Application Seeds (SYNTHETIC):** scripts/seed_data.py contains 10 mock farmers, 10 mock buyers, and 10 mock produce items used for frontend demonstration. In backend/dataset/historical_negotiations.json, 150 simulated negotiations exist.

6. 19,200 RECORD CLAIM VERIFICATION

Entity	Claimed Count	Actual Count in Repo	Verified Factual Status
BuyerAgent	19,200	14,078 clean / 13,179 features	CLAIM IS FALSE. Zero synthetic rows were generated to reach 19,200. Only authentic 14,078 APMC observations exist.
FarmerAgent	19,200	150 records	CLAIM IS FALSE / NOT FOUND. FarmerAgent has no 19,200 dataset; uses deterministic rule levels.

7. SEVEN-CROP VERIFICATION

Verified by running `scratch/verify_crop_implementation.py` against `shared/crop_catalog.py`:

Canonical Crop	Pricing Basis	Statutory Benchmark	Aliases Verified	Rejection Test
Sugarcane	FRP (CCEA)	FRP: █3.40/kg (MSP=None)	ganna, cane, sugar cane, oos	PASSED (17/17)
Soybean	MSP (CACP)	MSP: █48.92/kg	soya, soyabean, soyabeans	PASSED (17/17)
Cotton	MSP (CACP)	MSP: █71.21/kg	kapas, raw cotton, long staple	PASSED (17/17)
Jowar	MSP (CACP)	MSP: █33.71/kg	sorghum, jowari, jowar_hybrid	PASSED (17/17)
Onion	APMC Modal	Modal: █15.00–█26.00/kg (No MSP)	kanda, pyaz, onions	PASSED (17/17)
Bajra	MSP (CACP)	MSP: █26.25/kg	pearl millet, cumbu, bajri	PASSED (17/17)
Rice	MSP (CACP)	MSP: █23.00/kg	paddy, dhan, chawal, unhusked	PASSED (17/17)

8, 9 & 10. ML MODEL ARTIFACT & REPRODUCIBILITY VERIFICATION

Metric / Parameter	Claimed Value	Re-calculated Value	Verification Status
Model Artifact Path	<code>backend/models/buyer_price_prediction_model.pkl</code>	Exists (1,807 bytes)	VERIFIED
Model Pipeline Object	<code>Pipeline(StandardScaler, Ridge)</code>	<code>Pipeline(StandardScaler, Ridge(alpha=10.0))</code>	VERIFIED
Training Records	9,225 (70% chronological split)	9,225 records (`2014-09` to `2016-03`)	VERIFIED
Test Records	1,977 (15% chronological split)	1,977 records (`2016-07` to `2016-10`)	VERIFIED
Persistence Baseline MAE	█1.7258/kg	█1.7258/kg	100% REPRODUCIBLE
Moving Average (3p) MAE	█2.3949/kg	█2.3949/kg	100% REPRODUCIBLE
Ridge Regression MAE	█2.1812/kg	█2.1812/kg	100% REPRODUCIBLE
Beats Persistence?	NO (Persistence is strong)	NO (+█0.4554 difference)	VERIFIED
Beats Moving Average?	YES	YES (+█0.2137 better)	VERIFIED
Runtime Invocation	N/A	Uncalled in <code>`BuyerAgent.make_offer()`</code>	NOT IMPLEMENTED

11 & 12. RUNTIME ML & LIVE MARKET DATA VERIFICATION

- **Runtime ML Call Verification:** Traced `agents/buyer_agent.py` lines 1-830. `BuyerAgent` does **NOT** import or unpickle `buyer_price_prediction_model.pkl`. `make_offer()` computes bids using `self.target_price` and `self.reservation_price` passed at instantiation. The ML model is a verified offline artifact, but is **NOT WIRED AT RUNTIME**.
- **Live Market Data Probe Verification:** Directly queried `https://api.data.gov.in/resource/9ef84268...` via `scratch/verify_live_api.py` at 2026-09-14T20:59:05Z. The endpoint responded in 1.26 seconds with **HTTP 429: Too Many Requests** (`{"error": "Rate limit exceeded"}`). The live pipeline exists in code, but is currently rate-limited on the public gateway key.

13 & 14. LANGGRAPH & TEST SUITE VERIFICATION

Test Suite	Test Command	Total	Passes	Fail / Err	Verified Result
Buyer Extensive	python -m unittest tests/test_05_buyer_agent_extensive.py	38	38	0	PASSED (100%)
Buyer Economic State	python -m unittest tests/test_05_buyer_profile_economic_state.py	12	12	0	PASSED (100%)
7-Crop Isolation	python -m unittest tests/test_05_buyer_crop_isolation.py	17	17	0	PASSED (100%)
Data Ingestion & ML	python -m unittest tests/test_05_buyer_real_data_ingestion.py	7	7	0	PASSED (100%)
TOTAL BUYER TESTS	4 dedicated suites	74	74	0	74/74 PASSED (100%)
LangGraph Nodes	pytest tests/test_05_langgraph_nodes.py	29	25	4 failed	PARTIAL (NoneType bugs in graph)
FastAPI Marketplace	python -m unittest tests/test_marketplace_requirements.py	0 ran	0	1 error	Starlette TestClient kwargs bug

15. CLAIM VS REALITY MATRIX

CLAIM	VERIFIED?	EVIDENCE SOURCE	ACTUAL REALITY / FINDING
62,429 raw records	VERIFIED	Monthly_data_cmo.csv	Exactly 62,429 rows, 11 cols, 4.88 MB from MSAMB open data archive
14,078 clean records	VERIFIED	clean_buyer_market_data.csv	Exactly 14,078 rows, 10 cols, strictly 7 crops, 0 duplicates, 0 nulls
13,179 feature records	VERIFIED	buyer_feature_dataset.csv	Exactly 13,179 rows, 18 cols with forward target price and rolling features
Real agricultural data	VERIFIED	Inspection of APMC observations	100% genuine APMC mandi transactions across 349 Maharashtra mandis
7-Crop Isolation	VERIFIED	shared/crop_catalog.py	Sugarcane (FRP), Soybean, Cotton, Jowar, Onion, Bajra, Rice; all others rejected
74/74 Buyer tests pass	VERIFIED	4 unittest suites executed	74/74 tests passed in 1.8 seconds (100% pass rate)
Buyer ML Model Trained	VERIFIED	buyer_price_prediction_model.pkl	Ridge Regression trained on 9,225 records; metrics 100% reproducible
Buyer ML Runtime Used	FALSE	agents/buyer_agent.py lines 1-830	Model exists on disk, but BuyerAgent.make_offer() does NOT call it
Live market data works	PARTIALLY	Live HTTP probe to data.gov.in	Endpoint responds in 1.26s, but is rate-limited (HTTP 429)
LangGraph Integration	PARTIALLY	backend/agents/graph_orchestrator.py	Node 5 wired; 25/29 node tests pass; multi-agent simulation has state bugs
19,200 Buyer records	FALSE	Full repository file & grep scan	Does not exist; Buyer has 14,078 real rows (no synthetic padding)
19,200 Farmer records	FALSE	Full repository file & grep scan	Does not exist; Farmer has 150 simulated records in historical_negotiations.json

16. FINAL VERIFICATION VERDICT

- 7-Crop Configuration & Alias Normalization: VERIFIED.** Implemented in shared/crop_catalog.py and tested by 17 unit tests.
- Real Agricultural Dataset: VERIFIED.** 62,429 raw records, 14,078 clean records, and 13,179 feature rows verified by direct file parsing.
- Data Provenance: VERIFIED AS THIRD-PARTY MIRROR.** Originates from MSAMB Maharashtra government open data, retrieved via GitHub mirror.
- Zero Synthetic Data: VERIFIED.** No synthetic market data was generated for the BuyerAgent.
- 19,200 Record Claims: FALSE / CONTRADICTED.** Does not exist for Buyer (14,078 real) or Farmer (150 simulated).
- ML Model Artifact & Metrics: VERIFIED.** Ridge model artifact on disk; test MAE 2.1812 is 100% reproducible.
- Buyer Runtime ML Calling: NOT IMPLEMENTED.** The trained model is not yet connected to BuyerAgent.make_offer().
- Live Market Data: PARTIALLY VERIFIED.** Endpoint active, but currently blocked by public API gateway rate limits (HTTP 429).
- LangGraph Buyer Integration: PARTIALLY VERIFIED.** Node 5 exists and 25/29 tests pass; multi-agent simulation fails on NoneType.
- Git Safety & FarmerAgent Integrity: VERIFIED.** 0 lines modified in FarmerAgent; 100% clean isolation on feature/buyer-agent-verification.

END OF VERIFICATION REPORT — ALL FINDINGS PROVEN FROM PHYSICAL REPOSITORY ARTIFACTS